

Stat 5200, Module 4: Factorial Design and Analysis

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Overview

Objectives and overview

- This material is based on Chapter 8 of the textbook “A First Course in Design and Analysis of Experiments” (Oehlert, 2000).
- Module outcomes include:
 - You will understand complete and incomplete factorial treatment structures.
 - You will develop an intuitive and formal understanding of **interaction effects**.
 - You will be able to specify and fit additive and non-additive two-way ANOVA models.
 - You will learn how to construct and interpret main effect contrasts and interaction contrasts.
 - You will master the model testing sequence (testing interaction first).

Objectives and overview II

- You will know how to fit and analyze factorial designs in R using `aov()` and `emmeans()`.

Factorial Treatment Structure

Introduction to Factorial Structures

- In many experiments, we want to evaluate the effect of **two or more explanatory factors** simultaneously.
- A **factorial treatment structure** defines treatments as combinations of levels of each factor.
- Suppose Factor A has a levels and Factor B has b levels.
- A **complete factorial design** includes all $g = a \times b$ possible treatment combinations.

Example 1: 2×3 Diet Experiment

- Factor 1: Protein Source (Beef, Cereal, Pork) $\implies b = 3$.
- Factor 2: Protein Intake (High, Low) $\implies a = 2$.
- Total treatment combinations: $g = 2 \times 3 = 6$ unique diets.

Introduction to Factorial Structures II

Example 2: 2×2 Fertilizer Experiment

- Factor A: Superphosphate (P) $\rightarrow$ Levels: {Not Added, Added}
- Factor B: Potash (K) $\rightarrow$ Levels: {Not Added, Added}
- $g = 2 \times 2 = 4$ treatments:
 1. None (P_1K_1)
 2. Superphosphate only (P_2K_1)
 3. Potash only (P_1K_2)
 4. Both Superphosphate and Potash (P_2K_2)

Notation for Two-Way Factorials

- To identify observations in a two-way design, we use three subscripts:

$$y_{ijk}$$

- $i = 1, \dots, a$: level of Factor A
- $j = 1, \dots, b$: level of Factor B
- $k = 1, \dots, n$: replication number within treatment combination (i, j)
- Summary Notation:
 - Sample mean for treatment (i, j) : $\bar{y}_{ij\bullet} = \frac{1}{n} \sum_{k=1}^n y_{ijk}$
 - Marginal mean for level i of Factor A:
$$\bar{y}_{i\bullet\bullet} = \frac{1}{bn} \sum_{j=1}^b \sum_{k=1}^n y_{ijk}$$

Notation for Two-Way Factorials II

- Marginal mean for level j of Factor B:

$$\bar{y}_{\bullet j \bullet} = \frac{1}{an} \sum_{i=1}^a \sum_{k=1}^n y_{ijk}$$

- Overall grand mean: $\bar{y}_{\bullet \bullet \bullet} = \frac{1}{abn} \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n y_{ijk}$

Understanding Interaction Effects

What is an Interaction Effect?

- **Definition:** An interaction effect occurs when the effect of one factor on the response depends on the level of another factor.
- Ask yourself: *"Does the difference between levels of Factor A stay the same regardless of Factor B?"*
 - If **YES**: The factors are **additive** (No Interaction).
 - If **NO**: There is an **interaction** between factors.

What is an Interaction Effect? II

Intuition: "Difference of Differences"

Consider a 2×2 design with population means μ_{ij} :

- Simple main effect of A when Factor B is at level 1: $\mu_{11} - \mu_{21}$
- Simple main effect of A when Factor B is at level 2: $\mu_{12} - \mu_{22}$

The **interaction effect** is the difference between these two simple effects:

$$(\mu_{11} - \mu_{21}) - (\mu_{12} - \mu_{22})$$

If this difference equals zero, the effect of A is constant across levels of B. If non-zero, an interaction exists.

Graphical Diagnosis: Interaction Plots

- An **interaction plot** displays treatment group means across levels of one factor, with lines connecting levels of the second factor.
- **Parallel lines** $\implies$ No Interaction (Additive).
- **Non-parallel or crossing lines** $\implies$ Interaction present.

The Two-Way ANOVA Model

Model Specification

- We model the response y_{ijk} as:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where:

- μ : Overall grand mean.
- α_i : Main effect of level i of Factor A.
- β_j : Main effect of level j of Factor B.
- $(\alpha\beta)_{ij}$: Interaction effect between level i of A and level j of B.
- $\epsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2)$: Random errors.

Model Specification II

- Parameter Identification Restrictions (Sum-to-Zero Constraints):

$$\sum_{i=1}^a \alpha_i = 0, \quad \sum_{j=1}^b \beta_j = 0, \quad \sum_{i=1}^a (\alpha\beta)_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^b (\alpha\beta)_{ij} = 0$$

- Hierarchy of Models:

- Full (Non-additive) Model:**

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

- Additive Model:** $y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk}$

- Main Effect A Model:** $y_{ijk} = \mu + \alpha_i + \epsilon_{ijk}$

- Constant Model:** $y_{ijk} = \mu + \epsilon_{ijk}$

Factorial Contrasts

Main Effects vs. Interaction Contrasts

- A contrast in a factorial design is a linear combination of treatment means $\sum_i \sum_j w_{ij} \mu_{ij}$ subject to $\sum_i \sum_j w_{ij} = 0$.
- **Main Effect Contrasts for A:** Weights for a given level of A are identical across all levels of B.

$$\sum_{i=1}^a w_i = 0, \quad w_{ij} = w_i \quad \text{for all } j$$

Main Effects vs. Interaction Contrasts II

- **Interaction Contrasts ("Contrast of Contrasts")**: Weights are constructed as products of main effect contrast weights:
 $w_{ij} = w_{A,i} \times w_{B,j}$. Row sums and column sums of weights must **all equal zero**:

$$\sum_{i=1}^a w_{ij} = 0 \text{ for all } j, \quad \text{and} \quad \sum_{j=1}^b w_{ij} = 0 \text{ for all } i$$

Main Effects vs. Interaction Contrasts III

Example: 2×2 Factorial Weight Matrices

Main Effect A

(W_A)

1	1
-1	-1

Main Effect B

(W_B)

1	-1
1	-1

Interaction (W_{AB})

1	-1
-1	1

```
# Create factorial dataset
```

```
fert_data <- data.frame(  
  P = factor(rep(c("No_P", "P"), each = 8)),  
  K = factor(rep(rep(c("No_K", "K"), each = 4), 2)),  
  yield = c(183, 176, 291, 254,      # None  
            224, 258, 244, 214,      # K only  
            356, 300, 301, 271,      # P only  
            329, 283, 308, 326)      # P and K  
)
```

```
# Display sample means across cell combinations
```

```
with(fert_data, tapply(yield, list(P, K), mean))
```

```
# Generate interaction plot
```

```
interaction.plot(x.factor = fert_data$P, trace.factor = fert_data$K,  
  response = fert_data$yield, fun = mean,  
  type = "b", pch = c(16, 17), col = c("darkred", "darkblue"),  
  xlab = "Superphosphate (P)", ylab = "Mean Yield")
```

```
trace.label = "Potash (K)")
```

```
# Fit Two-Way ANOVA Model with Interaction  
fert_mod <- aov(yield ~ P * K, data = fert_data)  
summary(fert_mod)
```

References and Acknowledgements

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.

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